1. Create a tomcat container on 8080 and deploy sample application in tomcat.

```
ubuntu@ip-10-0-28-130:~$ sudo docker pull tomcat:9.0-jdk17
9.0-jdk17: Pulling from library/tomcat
4f4fb700ef54: Pull complete
d51ae22b1f36: Pull complete
4e07a0f12b2c: Pull complete
06ad70e463aa: Pull complete
2e4cbaed45ca: Pull complete
70cedf7730e9: Pull complete
0527d089727d: Pull complete
9abf99b2c8aa: Pull complete
8e019246f0d0: Pull complete
88711e85bf22: Download complete
3fa4039d3bcc: Download complete
Digest: sha256:5f6fb595a79e729ac7e79179b80a2e54f684f5430361add7b43d7e42a0144499
Status: Downloaded newer image for tomcat:9.0-jdk17
docker.io/library/tomcat:9.0-jdk17
ubuntu@ip-10-0-28-130:~$ sudo docker run -d \
  --name tomcat8080 \
  -p 8080:8080 \
  tomcat:9.0-jdk17
f3c7100624243782d4ab09895df704394862d902d51f74064be19ab78b029d2d
ubuntu@ip-10-0-28-130:~$ sudo docker ps
CONTAINER ID   IMAGE             COMMAND             CREATED          STATUS          PORTS                                                  NAMES
f3c710062424   tomcat:9.0-jdk17   "catalina.sh run"   14 seconds ago   Up 14 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   tomcat8080
ubuntu@ip-10-0-28-130:~$
```

Docker-02 Task 1 Completed

2. Create volume and deploy tomcat container on port 8081.

```
ubuntu@ip-10-0-28-130:~$ sudo docker exec tomcat8080 \
  mkdir -p /usr/local/tomcat/webapps/ROOT
ubuntu@ip-10-0-28-130:~$ sudo docker exec tomcat8080 sh -c \
  'echo "<html><body><h1>Docker-02 Task 1 Completed</h1></body></html>" > /usr/local/tomcat/webapps/ROOT/index.html'
ubuntu@ip-10-0-28-130:~$ sudo docker volume create tomcat-data
tomcat-data
ubuntu@ip-10-0-28-130:~$ sudo docker run -d \
  --name tomcat8081 \
  -p 8081:8080 \
  -v tomcat-data:/usr/local/tomcat/webapps/ROOT \
  tomcat:9.0-jdk17
38b2b90e838ed35138b38610a57f59a53d70dafd7962d0df871457280f728174
ubuntu@ip-10-0-28-130:~$ sudo docker exec tomcat8081 sh -c \
'echo "<html><body><h1>Docker-02 Task 2 Completed</h1></body></html>" > /usr/local/tomcat/webapps/ROOT/index.html'
ubuntu@ip-10-0-28-130:~$ sudo docker ps
CONTAINER ID   IMAGE             COMMAND             CREATED          STATUS          PORTS                                                  NAMES
38b2b90e838e   tomcat:9.0-jdk17   "catalina.sh run"   33 seconds ago   Up 33 seconds   0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   tomcat8081
f3c710062424   tomcat:9.0-jdk17   "catalina.sh run"   7 minutes ago    Up 7 minutes    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   tomcat8080
ubuntu@ip-10-0-28-130:~$
```

Docker-02 Task 2 Completed

3. Limit the nginx container to 500 MB.

```
ubuntu@ip-10-0-28-130:~$ sudo docker run -d \
  --name nginx500 \
  -p 8082:80 \
  --memory=500m \
  --memory-swap=500m \
  nginx:latest
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
6a6757ab48b1: Pull complete
b19352c60515: Pull complete
ecc510c1e359: Pull complete
113d9139ada2: Pull complete
792a76c59dbd: Pull complete
c9ec797c69ed: Pull complete
188e9077f350: Pull complete
3a5a894bf2b2: Download complete
2b7d75aa4652: Download complete
Digest: sha256:332d0beea97cff8c0d5d27976c0778a4b00345b9d74c2feaaf8efde548314948
Status: Downloaded newer image for nginx:latest
3e6093cc978b68efd3a4016b7a75c4e14fc45821ea706ca9503b60c694378957
ubuntu@ip-10-0-28-130:~$ sudo docker inspect nginx500 --format '{{.HostConfig.Memory}}'
524288000
ubuntu@ip-10-0-28-130:~$ sudo docker stats nginx500 --no-stream
CONTAINER ID   NAME       CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O         PIDS
3e6093cc978b   nginx500   0.00%     13.08MiB / 500MiB   2.62%     726B / 126B   10.5MB / 12.3kB   3
ubuntu@ip-10-0-28-130:~$
```

4. Create a sample docker file using below instructions.
    1. Base module as amazonlinux:latest
    2. Maintainer you name
    3. Install nginx
    4. COPY one index.html file to image
    5. Expose on port 80
    6. Command to start the nginx container

```
Complete!
 ---> Removed intermediate container 421de80aea68
 ---> f55bd419dc86
Step 4/6 : COPY index.html /usr/share/nginx/html/index.html
 ---> 5b5399464ee5
Step 5/6 : EXPOSE 80
 ---> Running in 18d3e026a4e6
 ---> Removed intermediate container 18d3e026a4e6
 ---> 81fb5ef4537e
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
 ---> Running in d8ff9c01d409
 ---> Removed intermediate container d8ff9c01d409
 ---> ef036d5e0424
Successfully built ef036d5e0424
Successfully tagged my-nginx:1.0
ubuntu@ip-10-0-28-130:~/nginx-docker$ sudo docker images
                                                                    i Info →  U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
amazonlinux:latest   8ed3c0a99684        227MB         59.3MB
my-nginx:1.0         ef036d5e0424        688MB          234MB
nginx:latest         332d0beea97c        243MB         66.8MB   U
tomcat:9.0-jdk17     5f6fb595a79e        737MB          234MB   U
ubuntu@ip-10-0-28-130:~/nginx-docker$
```

5. Push image to Docker hub

```
ubuntu@ip-10-0-28-130:~/nginx-docker$ sudo docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: XZJB-RPFH
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…
failed waiting for authentication: Global rate limit exceeded
ubuntu@ip-10-0-28-130:~/nginx-docker$ sudo docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'

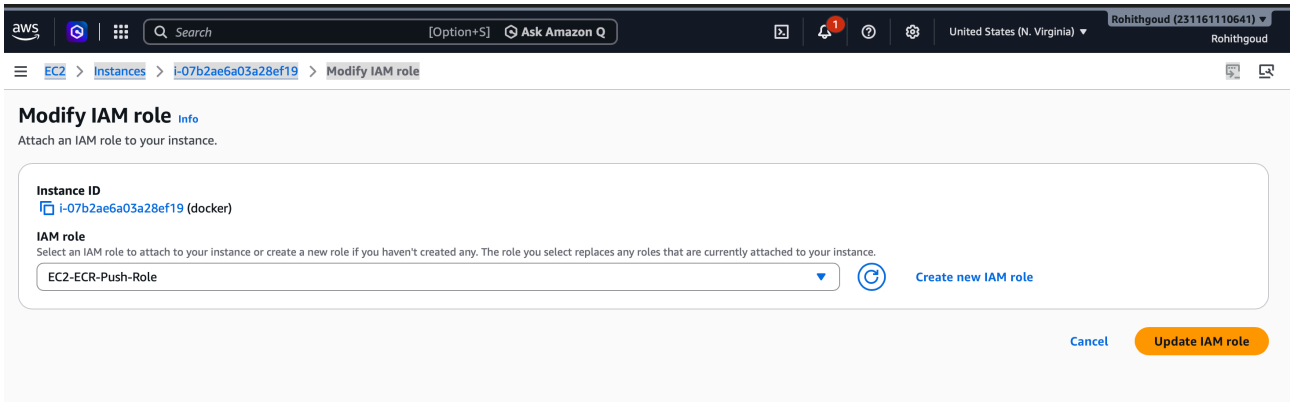

Your one-time device confirmation code is: ZPMD-GMGX
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-10-0-28-130:~/nginx-docker$ sudo docker tag my-nginx:1.0 rohithgoud15/my-nginx:1.0
ubuntu@ip-10-0-28-130:~/nginx-docker$ sudo docker push rohithgoud15/my-nginx:1.0
The push refers to repository [docker.io/rohithgoud15/my-nginx]
eb9dc979d2ee: Pushed
7234d0a9b6ba: Mounted from library/amazonlinux
67c051345ee2: Pushed
1.0: digest: sha256:ef036d5e04242e47f993b7aaa0cb542ebf6f7c2f87f4f3e17f40597544250aa7 size: 972
ubuntu@ip-10-0-28-130:~/nginx-docker$
```

6. push image to AWS ECR



Docker-Task-2

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Oct 15 13:41:40 2026 from 18.206.107.29
ubuntu@ip-10-0-28-130:~$ aws ecr create-repository --repository-name my-nginx --region us-east-1
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:us-east-1:231161110641:repository/my-nginx",
        "registryId": "231161110641",
        "repositoryName": "my-nginx",
        "repositoryUri": "231161110641.dkr.ecr.us-east-1.amazonaws.com/my-nginx",
        "createdAt": "2026-10-15T14:02:30.704000+00:00",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
ubuntu@ip-10-0-28-130:~$ aws ecr get-login-password --region us-east-1 | sudo docker login --username AWS --password-stdin 231161110641.dkr.ecr.us-east-1.amazona
ws.com

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-10-0-28-130:~$ sudo docker tag my-nginx:1.0 231161110641.dkr.ecr.us-east-1.amazonaws.com/my-nginx:1.0
ubuntu@ip-10-0-28-130:~$ sudo docker push 231161110641.dkr.ecr.us-east-1.amazonaws.com/my-nginx:1.0
The push refers to repository [231161110641.dkr.ecr.us-east-1.amazonaws.com/my-nginx]
eb9dc979d2ee: Pushed
7234d0a9b6ba: Pushed
67c051345ee2: Pushed
1.0: digest: sha256:ef036d5e04242e47f993b7aaa0cb542ebf6f7c2f87f4f3e17f40597544250aa7 size: 972
ubuntu@ip-10-0-28-130:~$
```

